Sets — Language, Representation, Subsets

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

LESSON 01 / 08

What a set is (and what it is not)

A **set** is a well-defined collection of distinct objects. Well-defined means membership is decidable: for any candidate x, the statement $x \in A$ is either true or false, never a matter of taste. The objects are called **elements** (or members). Distinct means a set lists each element once: $\{1, 1, 2\}$ means $\{1, 2\}$.

Notation. Capital letters (A, B, U) name sets, small letters (x, y, a) name elements. $x \in A$ reads "x is an element of A"; $x \notin A$ is its negation. The **cardinality** $n(A)$ (also written $|A|$) is the number of elements; it may be finite or infinite.

Three ways to write a set

- **Roster (listing):** $A = \{1, 2, 3\}$ — order and repetition do not matter.
- **Set-builder:** $B = \{x : x \text{ is an even prime}\} = \{2\}$. Read $\{x : P(x)\}$ as "the set of all x such that $P(x)$ holds".
- **Interval (only for subsets of $\mathbb{R}$, order matters):** (a, b) open, $[a, b]$ closed, $[a, b)$ or $(a, b]$ half-open, and rays $[a, \infty)$, $(-\infty, b)$. Infinity is never an element: always round brackets at $\pm\infty$.

ASSUMPTIONS & CONVENTIONS

Braces $\{\}$ always describe a set. $\in$ links an **element** to a set; $\subseteq$ links a **set** to a set. $1 \in \{1, 2\}$ but $\{1\} \subseteq \{1, 2\}$. Never mix the two levels: "$1 \subseteq \{1, 2\}$" is a type error, not a false inequality. All sets in a discussion are assumed to live inside a fixed **universal set** U once complements appear.

Kinds of sets

- **Empty (null) set** $\varnothing$: the set with no elements. There is exactly **one** empty set (uniqueness: any two sets with no elements are equal by definition of equality). $n(\varnothing) = 0$.
- **Singleton** $\{a\}$: one element. Note $\{a\} \neq a$ — the set and its element are different kinds of objects.
- **Finite / infinite:** $n(A)$ is a natural number vs not. $n(\mathbb{N})$ is infinite (countably so; no JEE counting applies).
- **Equal sets:** $A = B \Leftrightarrow$ every element of A is in B and vice versa. The standard proof is **double inclusion**: prove $A \subseteq B$ and $B \subseteq A$ separately.
- **Equivalent sets:** same cardinality only. $\{1, 2\}$ and $\{3, 4\}$ are equivalent but not equal. Do not say "equivalent" when you mean "equal".

Subsets and the power set

$A \subseteq B$ iff every element of A is an element of B. We write $A \subset B$ (or $A \subsetneq B$) when $A \subseteq B$ with $A \neq B$ (**proper** subset). Vacuous truth: $\varnothing \subseteq A$ for every A — there is no element of $\varnothing$ that could fail the test.

SHORT DERIVATION

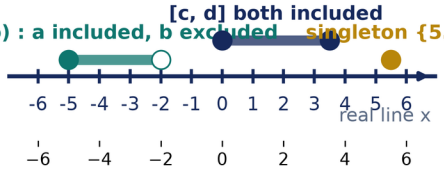
Why $|\mathcal{P}(A)| = 2^n$

Build a subset $S \subseteq A$ by deciding, for each of the n elements of A in turn, "in S" or "not in S". That is 2 choices per element, independently, so $2 \times 2 \times \cdots \times 2 = 2^n$ subsets. Small case $A = \{1, 2\}$: the four decisions list $\varnothing, \{1\}, \{2\}, \{1, 2\}$. Same reasoning: subsets with exactly k elements number $\binom{n}{k}$, so $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ (choosing each element is a subset of some size k).

KEY RESULT

The **power set** $\mathcal{P}(A)$ is the set of all subsets of A. Then $|\mathcal{P}(A)| = 2^{n(A)}$ and, for finite A: number of subsets $= 2^n$, number of proper subsets $= 2^n - 1$, number of non-empty subsets $= 2^n - 1$, number of non-empty proper subsets $= 2^n - 2$. Also $A \in \mathcal{P}(A)$ (as an element) and $A \subseteq A$ (as a subset) — two different statements, both true.

Interval conventions on the real line (filled dot = included)



The four confusions that cost marks

COMMON TRAP

1. $\emptyset$ **vs** $\{0\}$ **vs** $\{\emptyset\}$. $\emptyset$ has 0 elements. $\{0\}$ has one element (the number 0). $\{\emptyset\}$ has one element (the empty set). So $n(\{\emptyset\}) = 1$, and $\emptyset \in \{\emptyset\}$ is true while $\emptyset \in \{0\}$ is false. The set $\{\emptyset, \{0\}\}$ has exactly two elements.
2. $\in$ **vs** $\subseteq$. $\emptyset \subseteq \{0\}$ is true (subset), $\emptyset \in \{0\}$ is false (not an element). $0 \in \{0\}$ true, $\{0\} \in \{0\}$ false.
3. ***Equal vs equivalent.*** $\{1, 2\}$ and $\{2, 1\}$ are equal; $\{1, 2\}$ and $\{3, 4\}$ are only equivalent.
4. ***Two empty sets in one problem*** (e.g. $A \cap B = \emptyset$ vs $A \setminus B = \emptyset$) mean different things: the first says A, B are disjoint, the second says $A \subseteq B$.

WORKED EXAMPLE

Worked. $A = \{x \in \mathbb{Z} : |x| \le 2\}$ and $B = \{x \in \mathbb{Z} : 1 \le x \le 3\}$. Then $A = \{-2, -1, 0, 1, 2\}$, $B = \{1, 2, 3\}$, $A \cap B = \{1, 2\}$, $A \cup B = \{-2, -1, 0, 1, 2, 3\}$, $n(A \cup B) = 7$.
Check: the two listed sets share exactly $1, 2$, and every integer appearing in either list appears once in the union. Notice $|x| \le 2$ includes $x = -2$ — boundary cases always flip signs when absolute values appear.

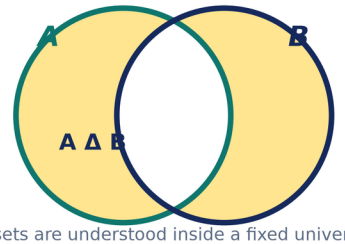
LESSON 02 / 08

The five operations

Fix a universal set U; all complements are relative to U.

- ***Union:*** $A \cup B = \{x : x \in A \text{ or } x \in B\}$ — "or" is inclusive; $A \cap B$ is counted in.
- ***Intersection:*** $A \cap B = \{x : x \in A \text{ and } x \in B\}$. Disjoint means $A \cap B = \emptyset$.
- ***Difference:*** $A \setminus B = \{x : x \in A \text{ and } x \notin B\} = A \cap B'$ — "in A, but not in B".
- ***Complement:*** $A' = U \setminus A = \{x \in U : x \notin A\}$. Without U there is no complement — always fix U first.
- ***Symmetric difference:*** $A \Delta B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$ — elements in ***exactly one*** of the two.

Shaded: A Δ B, the elements belonging to exactly one set



All sets are understood inside a fixed universe U

The algebraic laws (all provable by element-chasing)

KEY RESULT

Commutative: $A \cup B = B \cup A$, $A \cap B = B \cap A$, $A \Delta B = B \Delta A$. Associative: $(A \cup B) \cup C = A \cup (B \cup C)$ (same for $\cap$, Δ). Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. Idempotent: $A \cup A = A = A \cap A$. Complement: $A \cup A' = U$, $A \cap A' = \varnothing$, $(A')' = A$. Absorption: $A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$. Difference: $A \setminus B = A \cap B'$, $A \setminus A = \varnothing$.

SHORT DERIVATION

De Morgan's laws

$(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$. Chase elements: $x \in (A \cup B)'$ means "$x \notin A \cup B$", i.e. "$x \notin A$ ***and*** $x \notin B$" (negation flips "or" to "and"), i.e. $x \in A'$ and $x \in B'$, i.e. $x \in A' \cap B'$. Swapping $\cup$, $\cap$ and negating again gives the second law. The same argument with n sets gives: complement of a union is the intersection of complements, and vice versa — "keep the elements outside ***all*** of them" $= \bigcap A_i'$, "outside at least one" $= \bigcup A_i'$.

INTUITION

Read the operations as filters on a Venn diagram. $\cup$ = at least one condition true. $\cap$ = all conditions true. $\setminus$ = first true, second false. Δ = odd number of memberships. Every identity becomes "the shaded region is the same picture on both sides" — a genuine proof method once you say which region each element lands in.

Counting with a Venn diagram

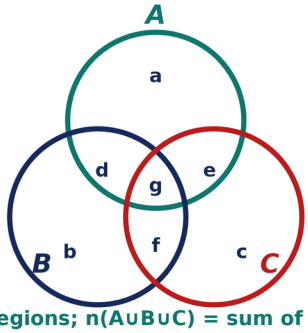
SHORT DERIVATION

Inclusion–exclusion

Partition $A \cup B$ into the three disjoint pieces $A \setminus B$, $A \cap B$, $B \setminus A$. Then $n(A \cup B) = n(A) + n(B) - n(A \cap B)$, because adding $n(A) + n(B)$ counts the overlap twice. For three sets (same idea, seven disjoint regions):

$$n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$$

where $\sum n(A)$ sums the three singletons and $\sum n(A \cap B)$ the three pairwise intersections.



Seven disjoint regions; n(A∪B∪C) = sum of all seven counts

WORKED EXAMPLE

Standard application. Of 60 students, 25 play cricket, 20 football, 15 both. How many play neither? $n(C \cup F) = 25 + 20 - 15 = 30$, so $n((C \cup F)') = 60 - 30 = 30$ students play neither. ***Check:*** the 15 who play both are inside both counts; subtracting once leaves each student counted exactly once. Boundary habit: "neither" is a complement, so the total 60 must be the universe.

COMMON TRAP

- A complement is meaningless without a stated universe — "the set of students not playing cricket" lives inside the class, not inside all humans.
- $A \setminus B = \varnothing$ means $A \subseteq B$, **not** $A = B$.
- $(A \setminus B) \cup B = A \cup B$ in general, not A. The reverse difference $B \setminus A$ is a different animal.
- Δ is not associative with $\cup$ mixing: parentheses matter when three operations combine.
- $n(A \cup B) = n(A) + n(B)$ holds **only** when $A \cap B = \varnothing$. Using it without checking disjointness is the classic counting error.

LESSON 03 / 08

Cartesian product of sets

An **ordered pair** (a, b) remembers order: $(a, b) = (c, d) \Leftrightarrow a = c$ and $b = d$ (this is the working convention; the formal set-theoretic build $(a, b) := \{\{a\}, \{a, b\}\}$ is a background fact, not an exam tool).

KEY RESULT

$A \times B = \{(a, b) : a \in A,\ b \in B\}$. Then $n(A \times B) = n(A) \cdot n(B)$ — product rule: choose the first coordinate ($n(A)$ ways) then the second ($n(B)$ ways), independently. $A \times B = \varnothing \Leftrightarrow A = \varnothing$ or $B = \varnothing$. $A \times B \neq B \times A$ in general (they are equal only in trivial cases such as $A = B$ or one empty).

Distributive-style facts (all set identities carry over with $\times$):

- $A \times (B \cup C) = (A \times B) \cup (A \times C)$ and $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
- $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$ — intersect **coordinate-wise**.
- $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.
- $A \times B \subseteq C \times D \Leftrightarrow (A \subseteq C$ and $B \subseteq D)$ when all four are non-empty.

COMMON TRAP

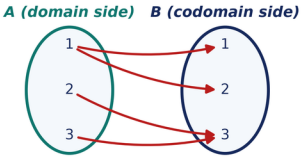
- $(A \times B) \times C \neq A \times (B \times C)$ as sets: the elements are $((a, b), c)$ vs $(a, (b, c))$. Only the elements are in natural bijection. JEE counts pairs, so keep parentheses as written.
- $A \times B$ is a **set of pairs**, not a number: $|A \times B|$ is the number; $A \times B$ itself contains points such as $(1, 4)$.
- $A \times A$ is the plane grid of a set with itself; $n(A \times A) = n(A)^2$, which is why an n-element set has n^2 ordered pairs — memorise this number, you will use it constantly.

Relations

A **relation** R from A to B is any subset $R \subseteq A \times B$. If $B = A$ we say "R is a relation on A". For $(a, b) \in R$ we often write $a\,R\,b$.

- **Domain** $= \{a \in A : \exists b,\ (a, b) \in R\}$ (first coordinates that actually appear).
- **Range** $= \{b \in B : \exists a,\ (a, b) \in R\}$.
- **Codomain** $= B$, fixed by declaration — range is a subset of the codomain, not necessarily equal to it.

Representations: roster form $R = \{(1, 1), (1, 2), (2, 3), (3, 3)\}$; rule form $\{(x, y) : y = x^2\}$; arrow diagrams; the 0-1 matrix ($m_{ij} = 1$ iff $(a_i, b_j) \in R$); graphs for relations on $\mathbb{R}$.



R = {(1,1),(1,2),(2,3),(3,3)}: a general relation — 1 sends two arrows, so R is not a function

Three properties of relations on a set A

- **Reflexive:** $a\,R\,a$ for **every** $a \in A$. One missing diagonal pair kills it.
- **Symmetric:** $a\,R\,b \Rightarrow b\,R\,a$ for all a, b.
- **Transitive:** $a\,R\,b$ and $b\,R\,c \Rightarrow a\,R\,c$.

Standard test cases on $\mathbb{Z}^+$ or $\mathbb{R}$: "$\leq$" is reflexive + transitive, **not** symmetric. "$\neq$" is symmetric, **not** reflexive, **not** transitive ($1 \neq 2$, $2 \neq 1$ but $1 \neq 1$ is false — the transitive conclusion fails). "divides" on $\mathbb{Z}^+$: reflexive + transitive, not symmetric ($2 \mid 4$ but $4 \nmid 2$). Parallelism of lines in a plane (with every line parallel to itself) is an equivalence relation.

SHORT DERIVATION

How many relations exist?

A relation on A is any subset of $A \times A$, and $n(A \times A) = n^2$ for $n = n(A)$. Each of the n^2 pairs is independently in or out of R: total relations $= 2^{n^2}$. Reflexive relations must contain all n diagonal pairs (a, a) and are free on the other $n^2 - n$ pairs: 2^{n^2-n}. Small case $n = 2$, $A = \{1, 2\}$: 4 pairs, $2^4 = 16$ relations; reflexive ones must contain $(1, 1), (2, 2)$ and are choices on $(1, 2), (2, 1)$: $2^2 = 4$ — list them: $\{(1, 1), (2, 2)\}$, plus each of $(1, 2)$ and/or $(2, 1)$.

WORKED EXAMPLE

Worked. $A = \{1, 2, 3\}$, $B = \{4, 5\}$. Then $A \times B$ has 6 pairs, so there are $2^6 = 64$ relations from A to B. Let $R = \{(x, y) \in A \times B : y - x \geq 3\}$. Check all six pairs: $(1, 4) : 4 - 1 = 3 \geq 3$ ✓; $(1, 5) : 4 \geq 3$ ✓; $(2, 4) : 2 \geq 3$ ✗; $(2, 5) : 3 \geq 3$ ✓; $(3, 4) : 1 \geq 3$ ✗; $(3, 5) : 2 \geq 3$ ✗. So $R = \{(1, 4), (1, 5), (2, 5)\}$, domain $= \{1, 2\}$, range $= \{4, 5\}$, codomain $= \{4, 5\}$. **Check:** the boundary $y - x = 3$ is included ($\geq$, not $>$), which is what makes $(1, 4)$ and $(2, 5)$ members — boundary conditions decide membership more often than bulk algebra.

LESSON 04 / 08

Equivalence relation

A relation R on A is an **equivalence relation** if it is reflexive, symmetric **and** transitive. The model example is congruence: for fixed $n \geq 2$, define $x\,R\,y \Leftrightarrow n \mid (x - y)$ ("$x \equiv y \pmod n$").

SHORT DERIVATION

Congruence mod n is an equivalence relation on $\mathbb{Z}$

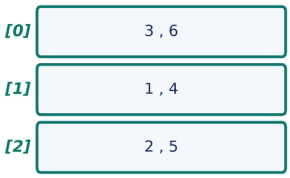
Reflexive: $x - x = 0$ and $n \mid 0$. Symmetric: if $n \mid (x - y)$ then $x - y = nk$, so $y - x = n(-k)$ and $n \mid (y - x)$. Transitive: $n \mid (x - y)$ and $n \mid (y - z)$ give $x - y = nk$, $y - z = nl$, so $x - z = n(k + l)$. Each step is one divisibility fact — nothing clever, which is exactly why this template reappears in exams.

Equivalence classes and partitions

For $a \in A$ the **class** $[a] = \{x \in A : x\,R\,a\}$. Fundamental facts: $a \in [a]$ (reflexivity); $[a] = [b] \Leftrightarrow a\,R\,b$; distinct classes are disjoint; the classes cover A. So the classes form a **partition** of A: a family of non-empty, pairwise disjoint blocks whose union is A.

KEY RESULT

Equivalence relations on A and partitions of A are the same data in two languages. $a\,R\,b \Leftrightarrow$ "a and b are in the same block". Counting either one gives the **Bell numbers**: $B_1 = 1$, $B_2 = 2$, $B_3 = 5$, $B_4 = 15$, $B_5 = 52$ — the number of equivalence relations on an n-element set. For $A = \{1, 2, 3\}$ the five partitions are: $\{\{1, 2, 3\}\}$; $\{\{1\}, \{2, 3\}\}$, $\{\{2\}, \{1, 3\}\}$, $\{\{3\}, \{1, 2\}\}$; $\{\{1\}, \{2\}, \{3\}\}$.



Partition induced by congruence mod 3 on {1,...,6}: three classes, one block each

WORKED EXAMPLE

Worked. On $\mathbb{Z}$ let $x\,R\,y \Leftrightarrow x + y$ is even. Symmetric is immediate ($x + y = y + x$); reflexive: $x + x = 2x$ even; transitive: if $x + y$ and $y + z$ are even, then $(x + y) - (y + z) = x - z$ is even and $x + z = (x - z) + 2z$ is even. Classes: even numbers $[0] = \{\ldots, -2, 0, 2, \ldots\}$ and odds $[1] = \{\ldots, -1, 1, 3, \ldots\}$. **Check:** $[2] = [0]$ since $2 + 0$ is even — classes are named by any member, not only by the smallest. Note this R is the same equivalence as $x \equiv y \pmod 2$.

The traps that generate exam questions

COMMON TRAP

1. **Symmetric + transitive does not imply reflexive.** On $A = \{1, 2, 3\}$ the relation $R = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$ is symmetric and transitive but not reflexive ($3 \notin$ any pair with itself).
2. **The empty relation on a non-empty** A is symmetric and transitive (vacuously) but not reflexive. The **universal relation** $A \times A$ is an equivalence relation (one big block). The **identity relation** $\{(a, a) : a \in A\}$ is also one (all singleton blocks).
3. **To prove "equivalence"** check the three axioms separately on arbitrary elements; a single counterexample kills any axiom. Many partial proofs assume transitivity "by symmetry" — never valid.
4. **Class names:** $[1] = [4]$ under mod 3; the partition blocks are sets of elements, the labels are cosmetic.

How JEE counts these (derivable with the syllabus)

SHORT DERIVATION

Reflexive and symmetric relations on an n-set

Reflexive + symmetric: the n diagonal pairs are forced in; off-diagonal, (a, b) and (b, a) must be chosen together — in or out as a pair. There are $\frac{n(n-1)}{2}$ such unordered pairs, so the count is $2^{n(n-1)/2}$. Small case $n = 3$: $2^3 = 8$ relations — forced $\{(1, 1), (2, 2), (3, 3)\}$ plus any subset of $\{(1, 2), (2, 1)\}$, $\{(1, 3), (3, 1)\}$, $\{(2, 3), (3, 2$, listed as $2^3 = 8$ choices. Similarly: reflexive alone $= 2^{n^2 - n}$, symmetric alone $= 2^{n(n+1)/2}$ (diagonal free + pairs together), antisymmetric $= 2^n \cdot 3^{n(n-1)/2}$ (diagonal free; each off-diagonal pair: neither, only (a, b), or only (b, a)).

STANDARD APPLICATION

Standard application. Number of relations on $\{1,2,3\}$ that are reflexive and symmetric $= 2^3 = 8$; that are equivalence relations $= B_3 = 5$; that are reflexive $= 2^6 = 64$. The examiner tests whether you separate the three axioms' constraints — diagonal forced, pairs coupled, or fully free.

LESSON 05 / 08

A function is a relation with two guarantees

$f: A \to B$ is a relation $f \subseteq A \times B$ such that every $a \in A$ has **exactly one** $b \in B$ with $(a,b) \in f$. The two halves — existence (at least one image) and uniqueness (at most one) — are checked separately on exam questions. A is the **domain** (declared; it is part of the definition), B the **codomain**, and $f(A) = \{f(a) : a \in A\} \subseteq B$ the **range** (image).

- Arrow-diagram test: every element of A sends exactly one arrow out of B (an element of B may receive many or none).
- Graph test on $\mathbb{R}$: vertical line test (every vertical line hits the curve at most once) plus domain coverage.

Notation: $y = f(x)$; $f: x \mapsto x^2 - 1$; f and g are equal ($f = g$) iff same domain, same codomain and same values everywhere.

ASSUMPTIONS & CONVENTIONS

Statements like "find the domain of f" assume the **maximal** real domain unless a domain is specified: all real x for which the formula makes sense. Range is always inside the declared codomain; when a question says $f: \mathbb{R} \to \mathbb{R}$ and asks whether f is onto, the codomain being all of $\mathbb{R}$ is part of the data. Writing $f: A \to B$ with A the natural domain and B chosen larger than the range makes f "into" but not onto — same formula, different function.

Finding domains (the rule stack)

KEY RESULT

Apply every restriction simultaneously, then intersect:

1. Polynomial: all of $\mathbb{R}$. 2. Rational $p(x)/q(x)$: $q(x) \neq 0$. 3. Even root $\sqrt[n]{g(x)}$ (n even): $g(x) \geq 0$. 4. Logarithm $\log_b g(x)$: $g(x) > 0$, base $b > 0, b \neq 1$. 5. $\tan g(x)$: $g(x) \neq \frac{\pi}{2} + k\pi$. 6. $1/g(x)$: additionally $g(x) \neq 0$. 7. Composite $g \circ f$: $x \in \operatorname{dom} f$ **and** $f(x) \in \operatorname{dom} g$. 8. Real powers $g^{p/q}$ in lowest terms: if q even, need $g \geq 0$.

WORKED EXAMPLE

Worked (domain). $f(x) = \sqrt{x-1} + \dfrac{\sqrt{5-x}}{x-3}$.

Conditions: $x - 1 \geq 0 \Rightarrow x \geq 1$; $5 - x \geq 0 \Rightarrow x \leq 5$; $x - 3 \neq 0 \Rightarrow x \neq 3$. Domain $= [1,3) \cup (3,5]$. **Boundary checks:** $x = 1$: $f(1) = 0 + \frac{2}{-2} = -1$ — defined, so the bracket at 1 is closed. $x = 5$: $f(5) = 2 + 0$ — defined, closed. $x = 3$: division by zero — deleted even though both roots are fine there. Endpoint testing is not optional: open vs closed brackets are where partial marks live.

Finding ranges (the four techniques)

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

Finding ranges (the four techniques)

- **Complete the square:** $x^2-4x+7=(x-2)^2+3\geq 3$, so range on $\mathbb{R}$ is $[3,\infty)$; the vertex $(2,3)$ is the only point where the bound is met.
- **Monotone pieces:** a strictly decreasing f maps $[a,b]$ onto $[f(b),f(a)]$ (order flips!). On $[0,1]$, x^2-4x+7 decreases from 7 to 4: range $[4,7]$.
- **Solve for x:** write $y=f(x)$, solve for $x=g(y)$, and require a real solution: for $y=\dfrac{x^2+1}{x^2-1}$ with $x^2\neq 1$... cleaner classic below.
- **Sign / bound analysis:** $|f|$, $e^x>0$, $\sqrt{\cdot}\geq 0$ give one-sided bounds; combine with continuity (the image of an interval is an interval).

WORKED EXAMPLE

Worked (range by discriminant).

$f(x)=\dfrac{x^2}{x^2+1}$ on $\mathbb{R}$. Solve $y=\frac{x^2}{x^2+1}$ for x:
$y(x^2+1)=x^2\Rightarrow x^2(1-y)=y\Rightarrow x^2=\frac{y}{1-y}$.
Real x exists iff $\frac{y}{1-y}\geq 0$, i.e. $0\leq y<1$ (note $y=1$ is impossible: $1-y=0$ gives $0=1$). Range $=[0,1)$. **Check:** $f(0)=0$ attained; $f(x)\to 1$ as $|x|\to\infty$ but never equals 1 — the range is half-open exactly as the algebra says.

COMMON TRAP

- $\sqrt{x^2}=|x|$, not x — the root output is never negative.
- $e^{\ln x}=x$ only for $x>0$; $\ln(e^x)=x$ for all real x. Swapping them silently changes domains.
- Range of a formula **changes with the domain**: x^2 on $\mathbb{R}$ has range $[0,\infty)$; on $[-1,2]$ it has range $[0,4]$.
- $f(A)\cap f(B)\supseteq f(A\cap B)$ always, but equality can fail (e.g. $f(x)=x^2$, $A=\{-2\}$, $B=\{2\}$): $f(A\cap B)=\varnothing$ while $f(A)\cap f(B)=\{4\}$. The similar-looking $f(A\setminus B)$ vs $f(A)\setminus f(B)$ is even more fragile — draw arrows before trusting algebra.

Image and preimage (defined for every function)

For $S\subseteq A$ the **image** is $f(S)=\{f(x):x\in S\}$.
For $T\subseteq B$ the **preimage** is $f^{-1}(T)=\{x\in A:f(x)\in T\}$. The superscript -1 here does **not** require invertibility — $f^{-1}(T)$ is a set of inputs, and it can be empty ($f^{-1}(\{-5\})=\varnothing$ for $f(x)=x^2$).

KEY RESULT

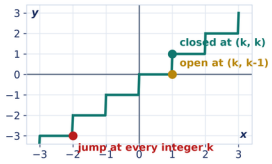
f^{-1} used as a **function** exists only for bijections (see the composition & inverse sheet). $f^{-1}(T)$ used as a **set** is always defined. Identities that always hold: $f(f^{-1}(T))\subseteq T$; $f^{-1}(f(S))\supseteq S$; $f^{-1}(T_1\cap T_2)=f^{-1}(T_1)\cap f^{-1}(T_2)$ (both directions); $f(S_1\cup S_2)=f(S_1)\cup f(S_2)$.

LESSON 06 / 08

The standard library (know the graphs cold)

- **Absolute value:** $|x| = x$ for $x \geq 0$, $-x$ for $x < 0$. Always ≥ 0; $|x| = \sqrt{x^2}$; triangle inequality $|a+b| \leq |a| + |b|$ with equality iff a, b have the same sign (or one is 0).
- **Signum:** $\operatorname{sgn}(x) = 1$ $(x > 0)$, 0 $(x = 0)$, -1 $(x < 0)$. So $x = |x|\operatorname{sgn}(x)$ for all x.
- **Greatest integer (floor):** $[x] = \lfloor x \rfloor$ = greatest integer $\leq x$. Fundamental sandwich: $[x] \leq x < [x] + 1$, equivalently $x = [x] + \{x\}$ with **fractional part** $\{x\} = x - [x] \in [0, 1)$.
- **Exponential / logarithm:** $e^x : \mathbb{R} \to (0, \infty)$ strictly increasing; $\ln x : (0, \infty) \to \mathbb{R}$ its inverse. $e^{\ln x} = x$ $(x > 0)$, $\ln e^x = x$ (all x).

Floor function y = [x]: right-continuous jumps at every integer (open circle = value excluded, closed = included)



COMMON TRAP

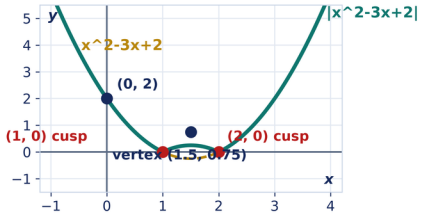
- $\sqrt{x^2} = |x|$, never x; $\sqrt{(-3)^2} = 3$, not -3.
- $[x]$ jumps by 1 at integers: $[0.999] = 0$ but $[1] = 1$. Draw open/closed dots explicitly — $[x]$ is **right-continuous**: value at the jump equals the right limit.
- $[x] + [-x] = 0$ if $x \in \mathbb{Z}$, else -1. $\{x\} + \{-x\} = 0$ if $x \in \mathbb{Z}$, else 1.
- $e^x > 0$ always: "$e^x = -2$" has no solution. $\ln x$ of a negative argument is undefined in $\mathbb{R}$.
- $|x| = x$ iff $x \geq 0$ (equality includes 0 — boundary!).

Graph transformations (one picture, many curves)

KEY RESULT

From $y = f(x)$: $f(x) + c$ shifts **up** c; $f(x + c)$ shifts **left** c (mind the sign); $af(x)$ scales vertically; $f(ax)$ scales horizontally by $1/|a|$ and reflects if $a < 0$; $|f(x)|$ reflects every negative lobe **upward** (creates cusps at old zeros); $f(|x|)$ keeps $x \geq 0$ and mirrors it into $x < 0$ (even function); $\{f(x)\}$ wraps each value into $[0, 1)$ (sawtooth along the curve); $1/f(x)$ turns zeros into vertical asymptotes and bounds into horizontal ones.

y = |x^2 - 3x + 2| (solid) from y = x^2 - 3x + 2 (dashed): roots at 1, 2 become cusps



WORKED EXAMPLE

Worked. Sketch $y = \{x\} = x - [x]$: straight segments of slope 1 from each integer up to, but not including, the next integer height 1; value 0 exactly at integers (closed dots at $(k, 0)$, open dots at $(k, 1)$). Range $[0, 1)$ — half-open, every time. Then $\{3.7\} = 0.7$, $\{-1.2\} = 0.8$ (because $[-1.2] = -2$).
Check: $\{x\} \in [0, 1)$ for negatives too — the floor of a negative number is the integer to its **left**, not "the integer part" of the digits.

STANDARD APPLICATION

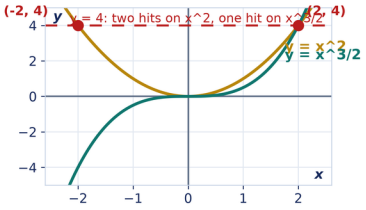
Standard application. Solve $|x - 1| + |x - 3| = 4$ by breaking at the knots $x = 1, 3$: on $x < 1$: $(1 - x) + (3 - x) = 4 \Rightarrow x = 0$ ✓; on $1 \leq x < 3$: $(x - 1) + (3 - x) = 2 \neq 4$; on $x \geq 3$: $(x - 1) + (x - 3) = 4 \Rightarrow x = 4$ ✓. Solution set $\{0, 4\}$. **Check boundaries:** $x = 1$ gives $0 + 2 = 2$; $x = 3$ gives $2 + 0 = 2$ — knots must be tested separately, they belong to exactly one case each.

LESSON 07 / 08

Injective, surjective, bijective

- **One-one (injective):** $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Test: horizontal line test on the graph; or solve $f(x_1) = f(x_2)$ and cancel. Strictly monotone (increasing or decreasing) on an interval ⇒ one-one there.
- **Onto (surjective):** range = codomain. "Into" just means not onto. To prove onto: for arbitrary y in the codomain, exhibit x with $f(x) = y$ and check x lies in the domain.
- **Bijection:** both. Exactly then does the inverse function f^{-1} exist (see the composition & inverse sheet).

Horizontal line test: y = x^2 is not one-one on R (line y = 4 crosses twice) but y = x^3 is



SHORT DERIVATION

How many functions A to B?

Each of the $n(A)$ inputs independently chooses one of the $n(B)$ outputs — product rule: $|B|^{|A|}$ functions. Among them: **injections** $= n(B)(n(B)-1)\cdots(n(B)-n(A)+1) = P(n(B$ if $n(A) \le n(B)$, else 0 (pigeonhole: more inputs than outputs forces a collision). **Bijections** $= n(A)!$ when $n(A) = n(B)$. **Surjections:** if $n(A) < n(B)$ none exist; otherwise count by cases for tiny sets — $f: \{1,2,3\} \to \{a,b\}$ has $2^3 = 8$ functions total, exactly 2 constant, so $8 - 2 = 6$ onto (each value must appear, i.e. not all three go to one side). The general inclusion-exclusion formula is Part II material; this case analysis is fully syllabus-level.

KEY RESULT

Small-case table for $|A| = 3, |B| = 2$: total functions $= 8$; one-one $= 0$; onto $= 6$; constant $= 2$. For $|A| = 2, |B| = 3$: total $= 9$; one-one $= 3 \cdot 2 = 6$; onto $= 0$; constant $= 3$. Memorise the shape of the table, not just the numbers — the constraints ($n(A) \le n(B)$ for injective, $n(A) \ge n(B)$ for surjective) are what get tested.

Even and odd functions

Requires a domain **symmetric about 0** (if x is allowed so is $-x$). Then f is **even** if $f(-x) = f(x)$ (graph symmetric about the y-axis), **odd** if $f(-x) = -f(x)$ (graph symmetric about the origin; rotation by 180°).

SHORT DERIVATION

Every function splits into even + odd parts

$f(x) = g(x) + h(x)$ where $g(x) = \frac{f(x)+f(-x)}{2}$ (even part) and $h(x) = \frac{f(x)-f(-x)}{2}$ (odd part). Check: $g(-x) = \frac{f(-x)+f(x)}{2} = g(x)$; $h(-x) = \frac{f(-x)-f(x)}{2} = -h(x)$. The split is unique. Consequences worth knowing: an odd f with 0 in its domain has $f(0) = 0$; even ± even = even, odd ± odd = odd; even × odd = odd; $f \cdot f$ is even whenever f is odd or even.

COMMON TRAP

- $f(x) = x^2$ on $[0,1]$ is **neither** even nor odd — the domain is not symmetric. Check the domain before the formula.
- $f(x) = x + 1$ is neither (sum of odd x and even 1 — sums of mixed parity are usually neither).
- $f(0) = 0$ is necessary for odd functions defined at 0, not sufficient ($f(x) = x^3 + x$ odd ✓, $f(x) = x^2$ has $f(0) = 0$ but is even).

Periodic functions

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

Periodic functions

f is **periodic** with period $T > 0$ if $f(x+T) = f(x)$ for all x (domain must allow $x+T$). The least such T is the **fundamental period** ("the period" in exam language).

KEY RESULT

Periods: $\sin x, \cos x$: 2π; $\tan x$: π; $\sin(ax+b), \cos(ax+b)$: $\frac{2\pi}{|a|}$; $\tan(ax+b)$: $\frac{\pi}{|a|}$; $|\sin x|$, $\sin^2 x$, $\cos^2 x$: π (taking absolute value or squaring merges the two halves); $\lfloor x \rfloor$, $\{x\}$: 1; a constant function satisfies $f(x+T) = f(x)$ for every T — it is periodic but has **no** fundamental period.

WORKED EXAMPLE

Worked. Period of $f(x) = \sin\left(\frac{x}{2}\right) + \cos(3x)$. First term: $\frac{2\pi}{1/2} = 4\pi$. Second: $\frac{2\pi}{3}$. Candidate period $= \text{lcm}(4\pi, \frac{2\pi}{3}) = 4\pi$ (need $T = k \cdot 4\pi = m \cdot \frac{2\pi}{3}$; least positive is 4π).

Check:
$f(x+4\pi) = \sin(\frac{x}{2} + 2\pi) + \cos(3x + 12\pi) = f($.
✓. Trap: lcms only exist for commensurable periods ($T_1/T_2 \in \mathbb{Q}$); $\sin x + \sin(\pi x)$ is periodic only in trivial senses — never assume a combined period exists.

STANDARD APPLICATION

Standard application. $f(x) = \sin^2\left(\frac{x}{2}\right)$: use $\sin^2\theta = \frac{1-\cos 2\theta}{2} = \frac{1-\cos x}{2}$, whose period is 2π — while naive reading suggests $\frac{2\pi}{1/2} = 4\pi$ for $\sin(x/2)$ then squaring halves it to 2π. Reduce with identities first, then read off the period; the two-stage guess 4π is wrong.

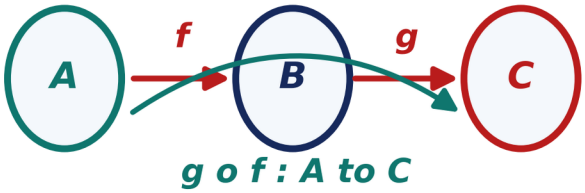
LESSON 08 / 08

Composition: $g \circ f$

$(g \circ f)(x) = g(f(x))$ — apply f **first**, then g.

KEY RESULT

Domain of $g \circ f$: $\{x \in \text{dom}\, f : f(x) \in \text{dom}\, g\}$. Order matters: $g \circ f \neq f \circ g$ in general. Composition is associative ($(h \circ g) \circ f = h \circ (g \circ f)$) because both sides equal $h(g(f(x)))$. If $f: A \to B$ and $g: B \to C$ then $g \circ f: A \to C$ and $f(A) \subseteq B$ must land inside g's domain — if f's range escapes $\text{dom}\, g$, shrink f's domain until it does not.



WORKED EXAMPLE

Worked. $f(x) = \sqrt{x}$ (domain $[0, \infty)$), $g(x) = x - 3$ (domain $\mathbb{R}$). Then $(g \circ f)(x) = \sqrt{x} - 3$ with domain $[0, \infty)$ — every output of f is fine for g. But $(f \circ g)(x) = \sqrt{x-3}$ has domain $[3, \infty)$ — g must not leave f's domain. **Check:** $x = 0$: $g(f(0)) = -3$ defined; $f(g(0)) = \sqrt{-3}$ undefined. The same two functions give different composite domains; the direction of composition is the whole question.

Inverse functions

$f: A \to B$ has an **inverse** $f^{-1}: B \to A$ iff f is bijective; then $f^{-1}(f(x)) = x$ for $x \in A$ and $f(f^{-1}(y)) = y$ for $y \in B$. Recipe: write $y = f(x)$, solve for $x = g(y)$, swap letters: $f^{-1}(x) = g(x)$. The domain of f^{-1} is the **range** of f and vice versa.

SHORT DERIVATION

$(g \circ f)$ inverse rule

$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ — the socks-and-shoes law: undo g first, then f. Check: $(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ f = I$ ✓, and symmetrically on the other side (needed since $g \circ f$ bijective implies each of f, g suitably bijective between the right sets).

Graphically $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in the line $y = x$ (points $(a, b) \mapsto (b, a)$). If f is one-one but not onto, restrict the codomain to the range; if onto but not one-one, restrict the domain (e.g. x^2 on $[0, \infty)$ has inverse $\sqrt{x}$).

WORKED EXAMPLE

Worked. $f(x) = \frac{2x+3}{x-1}$, $x \neq 1$. Solve $y = \frac{2x+3}{x-1}$: $y(x-1) = 2x+3 \Rightarrow x(y-2) = y+3 \Rightarrow x = \frac{y+3}{y-2}$. So $f^{-1}(x) = \frac{x+3}{x-2}$, domain $x \neq 2$ = range of f. Vertical asymptote $x = 1$, horizontal asymptote $y = 2$ of f become the horizontal $y = 1$ and vertical $x = 2$ of f^{-1} — asymptotes swap roles under reflection in $y = x$, exactly like domain and range. **Check:** $f(f^{-1}(3)) = f(6) = \frac{15}{5} = 3$ ✓; $f^{-1}(f(2)) = f^{-1}(7) = \frac{10}{5} = 2$ ✓.

COMMON TRAP

- f^{-1} as an inverse **function** needs bijection; writing f^{-1} before checking one-one + onto is the commonest Chapter-1 error.
- $(f \circ g)^{-1} \neq f^{-1} \circ g^{-1}$ (order flips!). The square bracket notation $f^{-1}(x) \neq \frac{1}{f(x)}$ — the -1 superscript means inverse function (or preimage set), never reciprocal, except the (rare) $\sin^{-1}$ convention confusion: $\sin^{-1} x$ is the inverse sine, while $(\sin x)^{-1} = \csc x$.
- $f \circ g = g \circ f$ happens only in special cases (e.g. $f = g$; or both powers x^a, x^b; or $f = g^{-1}$ giving identity) — never assume it.

Chapter recap — the whole sheet in one place

KEY RESULT

Sets: $|\mathcal{P}(A)| = 2^n$; equal = double inclusion; $\emptyset, \{0\}, \{\emptyset\}$ distinct. Operations: De Morgan $(A \cup B)' = A' \cap B'$; $A \setminus B = A \cap B'$; $A \Delta B = (A \cup B) \setminus (A \cap B)$; $n(A \cup B) = n(A) + n(B) - n(A \cap B)$; three-set version with alternating signs. Products & relations: $n(A \times B) = n(A)n(B)$; relations on n-set $= 2^{n^2}$; reflexive $= 2^{n^2-n}$; symmetric $= 2^{n(n+1)/2}$; reflexive+symmetric $= 2^{n(n-1)/2}$; equivalence = partitions $= B_n$. Functions: $|B|^{|A|}$ total; injections $P(|B|, |A|)$; bijections $n!$; domain rule stack; $\text{dom}(g \circ f)$ careful; f invertible iff bijective; $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$; even/odd need symmetric domain; periods scale as $1/|a|$.

STANDARD APPLICATION

How JEE asks this chapter. (i) Domain/range of built formulas with roots, logs, floors — always test endpoints. (ii) Counting: number of relations/functions with axiom constraints. (iii) Equivalence classes: partition a concrete set, name classes. (iv) Function classification: one-one/onto/even/odd/periodic verdicts on a given formula. (v) Composite domains and inverse formulas with parameter conditions. Each is a two-line application of the boxes above — the marks go to boundary discipline, not to new algebra.

LESSON 01 / 03

Status of this sheet

Part I derived total, reflexive, symmetric and reflexive+symmetric counts with direct pair analysis. This sheet completes the catalogue (antisymmetric, equivalence/Bell, surjections by inclusion–exclusion) and shows the small-case listings an IOQM/RMO solution must carry. All counts are **models built from the Part I supply list** (product rule, complements, inclusion–exclusion on sets) — the technique is exam-derivable even where the topic is listed as an extension.

The catalogue for $|A| = n$

KEY RESULT

On an n-element set: relations $= 2^{n^2}$; reflexive $= 2^{n^2-n}$; symmetric $= 2^{n(n+1)/2}$; antisymmetric $= 2^n \cdot 3^{n(n-1)/2}$; reflexive+symmetric $= 2^{n(n-1)/2}$; reflexive+transitive = number of "preorder with fixed diagonal" (no simple closed form — casework); equivalence relations $= B_n$ (Bell): $1, 2, 5, 15, 52$ for $n = 1..5$. Functions $A \to B$ with $|B| = m$: m^n; injective: $\frac{m!}{(m-n)!}$ if $n \le m$ else 0; surjective: $n!\, S(n,m)$ with S the Stirling number $= \sum_{i=0}^{m} (-1)^i \binom{m}{i} (m-i)^n$.

SHORT DERIVATION

Surjections by inclusion-exclusion

$f : A \to B$ surjective iff no element of B is missed. Let U = all m^n functions and E_b = functions missing the value b. We want $|U \setminus \bigcup_b E_b|$. Now $|E_b| = (m-1)^n$ (values chosen from the rest), $|E_b \cap E_c| = (m-2)^n$, and so on, so inclusion–exclusion gives $\sum_{i=0}^{m} (-1)^i \binom{m}{i} (m-i)^n$. Small case $n = 3, m = 2$: $2^3 - 2 \cdot 1^3 = 6$ ✓ matches the 6 onto functions $\{1,2,3\} \to \{a,b\}$ found by cases in Part I. Sanity: $m > n$ gives 0 (e.g. $n = 2, m = 3$: $9 - 3 \cdot 4 + 3 \cdot 1 = 0$).

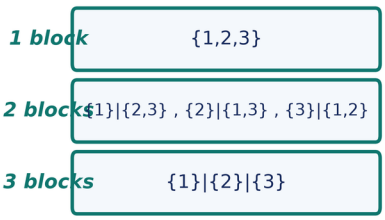
WORKED EXAMPLE

Worked. Onto functions $\{1,2,3,4\} \to \{a,b,c\}$: $3^4 - \binom{3}{1}2^4 + \binom{3}{2}1^4 - \binom{3}{3}0^4 = 81 - 48 + 3 =$. Cross-check with Stirling: $S(4,3) = 6$ (partitions of 4 elements into 3 blocks: one block of 2 and two singletons, $\binom{4}{2} = 6$), then order the blocks onto a, b, c: $6 \cdot 3! = 36$ ✓. Two independent methods agreeing is the standard olympiad "check" line.

Bell numbers by listings (do this in solutions)

For $A = \{1,2,3,4\}$, $B_4 = 15$ partitions: 1 block (1 way), 2 blocks: choose the block containing 1 ($S(4,2) = 7$), 3 blocks ($S(4,3) = 6$), 4 singletons (1). Total $1 + 7 + 6 + 1 = 15$ ✓.

General recurrence (extension): $B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k$ — element $n+1$ joins a block after k others are already placed... more cleanly: $B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_{n-k}$: choose which elements share the block of $n+1$.



The 5 equivalence relations on {1,2,3} as partitions (Bell B_3 = 5)

COMMON TRAP

- **Labelled vs unlabelled.** $\{\{1,2\},\{3,4\}\}$ and $\{\{1,3\},\{2,4\}\}$ are different partitions of a labelled set; counting unlabeled block *shapes* is a different (smaller) problem. State which one you count.
- Antisymmetric count $2^n 3^{n(n-1)/2}$: each off-diagonal unordered pair $\{a,b\}$ allows exactly 3 states — neither direction, only $a \to b$, only $b \to a$ — never 4 (both directions are banned).
- Inclusion–exclusion signs alternate. Boundary convention: a function on an empty domain is unique (the empty function), so $m^0 = 1$ for every m, including $0^0 = 1$ — state the convention whenever $n = 0$ appears.
- Surjections need $n \geq m$; injections need $n \leq m$. A formula that spits out a nonzero answer violating these is being used wrong.

ASSUMPTIONS & CONVENTIONS

Counts assume **finite** sets and labelled elements throughout. $S(n,m)$ here means Stirling numbers of the second kind (onto functions divided by $m!$); the notation is Part II vocabulary, but every *numerical* claim above is checkable by listing for $n, m \leq 4$ — an olympiad write-up should always include one such listing.

Status of this sheet

The equations below are **additional theory** (classic RMO/INMO and AIME-adjacent material). Nothing here contradicts the syllabus: every *model* shown is solvable with the Part I supply list plus one extra idea, stated openly. Do not claim such an equation "can never appear in JEE" — exam papers have historically leaked olympiad-flavoured models; what is *not* promised is any theorem beyond the statements given here.

The Cauchy equation $f(x+y) = f(x) + f(y)$ on $\mathbb{R}$

SHORT DERIVATION

Rational linearity

$f(0) = f(0+0) = f(0) + f(0) \Rightarrow f(0) = 0$.

Induction: $f(nx) = nf(x)$ for $n \in \mathbb{N}$; then $0 = f(0) = f(x + (-x)) = f(x) + f(-x) \Rightarrow f(-x)$, so $f(nx) = nf(x)$ for all integers n. For p/q with $q \geq 1$:

$f(p) = f\left(q \cdot \frac{p}{q}\right) = qf\left(\frac{p}{q}\right) \Rightarrow f\left(\frac{p}{q}\right) = \frac{p}{q} f(1)$. Hence $f(r) = cr$ for every rational r, where $c = f(1)$.

KEY RESULT

Regular upgrade. If f is additionally monotone on any interval, or continuous at a point, or bounded above/below on any interval, then $f(x) = cx$ for all real x (rational-linearity plus density of $\mathbb{Q}$ and the squeeze). Without *some* regularity, non-linear solutions exist (via a Hamel basis; they are non-measurable and cannot be written down explicitly) — so: **never write $f(x) = cx$ as "the" solution unless the problem grants a regularity condition.** Many exam models quietly assume continuity; state it.

LESSON 02 / 03

WORKED EXAMPLE

Worked models.

1. $f(x+y)=f(x)+f(y)$ for all $x,y\in\mathbb{Q}$, $f(1)=2$; then $f(7/3)=\frac{7}{3}\cdot 2=\frac{14}{3}$ — no regularity needed, because the domain is $\mathbb{Q}$ itself (rational linearity already complete).
2. $f:\mathbb{R}\to\mathbb{R}$ continuous, $f(x+y)=f(x)+f(y)$, $f(2)=6$; $c=f(1)=3$, so $f(x)=3x$ for every real x. **Check:** $f(x+y)=3(x+y)=3x+3y$ ✓.
3. $f:\mathbb{R}\to\mathbb{R}$ with $f(x+y)=f(x)+f(y)$ and $f(x)\geq 0$ for all $x>0$ (bounded below on $(0,\infty)$); the regularity condition is what upgrades $f(r)=cr$ to $f(x)=cx$; then $f(x)\geq 0$ for $x>0$ forces $c\geq 0$.

The multiplicative cousins

KEY RESULT

$f(x+y)=f(x)f(y)$: either $f\equiv 0$, or $f(0)=1$ and f never vanishes; with continuity $f(x)=a^x$, $a=f(1)>0$. $f(xy)=f(x)+f(y)$ on $(0,\infty)$: with continuity $f(x)=c\ln x$ (set $g(t)=f(e^t)$ and Cauchy reduces). $f(xy)=f(x)f(y)$ on $(0,\infty)$: $f(x)=x^c$ or $f\equiv 0,1$ specials. **Pexider:** $f(x+y)=g(x)+h(y)$ for all x,y implies all three are affine with $f=g+h+\text{const}$ relationships — plug $y=0$ to express g,h through one function and fall back to Cauchy. **Jensen:** $f\left(\frac{x+y}{2}\right)=\frac{f(x)+f(y)}{2}$ with continuity $\Rightarrow f(x)=cx+d$.

SHORT DERIVATION

$f(x+y)=f(x)f(y)$ quickly

Put $y=0$: $f(x)=f(x)f(0)$, so either $f\equiv 0$ or $f(0)=1$. If $f(x_0)=0$ for some x_0, then $f(x)=f((x-x_0)+x_0)=f(x-x_0)f(x_0)=0$ for all x: $f\equiv 0$. So a non-zero solution never vanishes; if it takes both signs it still can't cross 0 (continuous image of connected set), hence a positive solution has $f(x)=e^{g(x)}$ with g additive, and continuity finishes $g(x)=x\ln a$.

Involutions $f(f(x))=x$

$f\circ f=\mathrm{id}$ means f is its own inverse: f is bijective and $f^{-1}=f$. Solutions of $f(f(x))=x$ split into **fixed points** ($f(x)=x$) and **2-cycles** ($f(x)=y\neq x$, $f(y)=x$).

SHORT DERIVATION

Increasing involution fixes everything

Suppose f is strictly increasing and $f(f(x))=x$. If $f(x_0)>x_0$ then applying the increasing f gives $f(f(x_0))>f(x_0)>x_0$, i.e. $x_0>f(x_0)$ — contradiction. Similarly $f(x_0)<x_0$ contradicts. So $f(x)=x$ for all x: every increasing involution is the identity on its domain. A decreasing involution (e.g. $f(x)=a-x$, $f(x)=1/x$ on $\mathbb{R}^+$) instead swaps pairs around fixed points.

WORKED EXAMPLE

Worked. Find all continuous $f:\mathbb{R}\to\mathbb{R}$ with $f(f(x))=x$ and f decreasing. Any x with $f(x)=x$ is a fixed point; $g(x)=f(x)-x$ is decreasing (f dec, $-x$ dec... sum of two decreasing is decreasing) hence g has at most one zero — at most one fixed point. Example $f(x)=a-x$ has exactly one ($x=a/2$). The general decreasing-continuous involution picture: one fixed point (if g changes sign) and the rest of the line paired off in 2-cycles symmetric about it. **Check:** $f(x)=1/x$ on $(0,\infty)$: decreasing, $f(f(x))=x$ ✓, fixed point $x=1$ ✓.

COMMON TRAP

- Cauchy with domain $\mathbb{Q}$: rational linearity is the complete answer. Cauchy with domain $\mathbb{R}$: rational linearity + "regularity" is required; state the condition.
- $f(x)=x^2$ is even, $f(x)=x^3$ odd, but $f(x)=x^2+x^3$ is neither — decomposition (Part I) is the safe tool.
- In $f(f(x))=x$ problems, plugging $x=f(t)$ is the standard trick: $f(f(f(t)))=f(t)$, i.e. $f^3=f$; combined with $f^2=\mathrm{id}$ this just re-proves bijectivity — write the injection/surjection argument once, cleanly.
- Periodic + additive: $f(x+T)=f(x)$ with additivity forces $f(T)=f(0)=0$, and with continuity that collapses $f\equiv 0$ (since $f(r)=cr$ and $cT=0$). Without continuity a wild additive T-periodic solution can still exist on a Hamel basis — so again: state the regularity condition before claiming collapse. The point stands: ***imposing two strong structures usually collapses the family to trivial members, but "usually" needs a hypothesis.***

LESSON 03 / 03

Status of this sheet

The greatest-integer function is ***in the official syllabus*** (listed under special functions); Hermite's identity and the fixed-point decomposition are extra theory. Every worked model below is solved with syllabus tools only — so treat the identities as accelerators, and the models as fair game.

Hermite's identity

SHORT DERIVATION

Sum of n shifted floors

For real x and positive integer n:

$$\sum_{k=0}^{n-1}\left\lfloor x+\tfrac{k}{n}\right\rfloor=\lfloor nx\rfloor$$

Write $x=m+\theta$ with $m=\lfloor x\rfloor$, $\theta\in[0,1)$. Then $\lfloor x+\frac{k}{n}\rfloor=m+\lfloor\theta+\frac{k}{n}\rfloor$, so the left side is $nm+\sum_k\lfloor\theta+\frac{k}{n}\rfloor$. Now $\lfloor\theta+\frac{k}{n}\rfloor=1$ exactly when $\frac{k}{n}\ge 1-\theta$, i.e. $k\ge n(1-\theta)$; the number of $k\in\{0,\dots,n-1\}$ with $k\ge n(1-\theta)$ is $\lfloor n\theta\rfloor$ (boundary $n(1-\theta)$ an integer counts k from that value up). So the sum $=nm+\lfloor n\theta\rfloor=\lfloor nm+n\theta\rfloor=\lfloor nx\rfloor$ ✓.

WORKED EXAMPLE

Worked checks. $n=3$, $x=1.7$: $\lfloor 1.7\rfloor+\lfloor 1.7+0.33\rfloor+\lfloor 1.7+0.67\rfloor=1+2+$ ✓. $x=2$ exactly: $2+2+2=6=\lfloor 6\rfloor$ ✓.
Check boundary: when $n\theta$ is an integer the threshold $k\ge n(1-\theta)$ is attained (included) — $\lfloor\cdot\rfloor$ inequality is $\ge$ on both sides of the threshold count.

Floor identities and equations

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

Floor identities and equations

KEY RESULT

$[x] \le x < [x]+1$;
$[x]+[y] \le [x+y] \le [x]+[y]+1$;
$[x]+[-x] = 0$ *if* $x \in \mathbb{Z}$ *else* -1;
$\{x\}+\{-x\} = 0$ *or* 1 *with the same split;*
number of integers in $(a, b]$ *is* $[b]-[a]$;
$[x] = c \Leftrightarrow x \in [c, c+1)$; $[x] < c \Leftrightarrow x < c$ *(for integer* c*);* $[x] \ge c \Leftrightarrow x \ge c$ *(integer* c*).*

STANDARD APPLICATION

Model (fully syllabus-level). *Solve* $\lfloor x^2 \rfloor = 2$ *over the reals. Sandwich:* $2 \le x^2 < 3$*. Left half:* $x^2 \ge 2 \Rightarrow |x| \ge \sqrt{2}$*; right half:* $x^2 < 3 \Rightarrow |x| < \sqrt{3}$*. Solution set* $= \{x : \sqrt{2} \le |x| < \sqrt{3}\} = [-\sqrt{3}, -\sqrt{2}] \cup [\sqrt{2}$ *—* ***closed*** *at* $\pm\sqrt{2}$ *(*$x^2 = 2$ *gives* $\lfloor 2 \rfloor = 2$ ✓*) and* ***open*** *at* $\pm\sqrt{3}$ *(*$x^2 = 3$ *would give* $\lfloor 3 \rfloor = 3$ ✗*). The answer is a set of reals, not a number. Integer version:* $x \in \mathbb{Z}$ *needs* $x^2 \in [2, 3)$*, but squares of integers are* $0, 1, 4, \ldots$ *—* ***no solutions.*** ***Check:*** *the two phrasings ("all real* x*" vs "all integer* x*") have completely different answers; the quantifier is part of the question.*

Fixed points and involutions (counting flavour)

$f(x) = x$ *is a* ***fixed point****;* $f(f(x)) = x$ *is an* ***involution equation*** *whose solution set decomposes into fixed points of* f *plus 2-cycles* $\{x, f(x)\}$ *with* $f(x) \ne x$*.*

KEY RESULT

If f *is strictly increasing and* $f(f(x)) = x$ *on an interval, then* $f(x) = x$ *everywhere on it (Part II sheet 02 proof). If* f *is strictly decreasing and continuous,* $g(x) = f(x) - x$ *is strictly decreasing, so* f *has* ***at most one*** *fixed point; each non-fixed* x *pairs with* $f(x)$ *in a 2-cycle. Counting solutions of* $f(f(x)) = x$ *for a concrete monotone* f *therefore reduces to: find fixed points (solve* $f(x) = x$*), then account for the symmetric 2-cycles.*

WORKED EXAMPLE

Worked. $f(x) = 3 - x$ *on* $\mathbb{R}$*: decreasing,* $f(f(x)) = 3 - (3 - x) = x$ ✓ *for* ***every*** x *— the entire line solves the involution equation, yet only* $x = 3/2$ *is a fixed point. Moral: "solutions of* $f(f(x)) = x$*" and "fixed points of* f*" are different sets; the question wording decides which is asked. With* $f(x) = \frac{1}{x}$ *on* $(0, \infty)$*:* $f(f(x)) = x$ *everywhere; fixed point* $x = 1$ *only.* ***Check:*** $f(2) = 1/2$*,* $f(1/2) = 2$ *— a 2-cycle, as predicted by the decreasing-case analysis.*

COMMON TRAP

- $\lfloor x \rfloor$ of a negative number: $\lfloor -1.2 \rfloor = -2$, not -1 (greatest integer **less or equal**). Truncation toward zero is a different function.
- In floor equations, the answer set's endpoints are decided by strict vs non-strict inequalities — write the sandwich first, then convert to interval notation, then test both endpoints.
- $\lfloor x^2 \rfloor = 2$ vs $\lfloor x \rfloor^2 = 2$: the second means $\lfloor x \rfloor = \pm\sqrt{2}$, impossible for integer $\lfloor x \rfloor$ — no solutions. Composition order matters even inside floor equations.
- Never square an inequality $\sqrt{2} \le |x| < \sqrt{3}$ without tracking that $|x|$ is already non-negative — squaring is safe on $[0, \infty)$ but flips nothing here; for $x < 0$ reflect by $|x|$.

ASSUMPTIONS & CONVENTIONS

All floor statements are for real x; n is a positive integer in Hermite; fixed-point claims assume the stated monotonicity/continuity on an interval. 2-cycles are unordered pairs $\{x, y\}$ with $y = f(x) \neq x$; when "count solutions" is asked on a finite set, count elements x, not cycles.